

Matrix functions and stability

Imaginary-part rank in a limiting Green-function matrix equation

Difficulty: challenging **Importance:** interesting to specialist**Status:** Partially resolved

Rating rationale: Challenging because the limiting complex solution must be tied exactly to unit-circle eigenstructure; specialist impact is on structured Green-function matrix equations.

Auxiliary result; canonical target unchanged — 2026-09-11

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For real A, Q with $Q = Q^T$, scalar regularization $i\eta I$, and a finite invertible stabilizing limit, the manuscript proves that the imaginary-part rank is half the number of odd unit-circle Jordan blocks. It also establishes semisimple regularity and an exact defective example.

This does not settle the canonical general complex C, D, R, P problem. Its simple-eigenvalue real subcase was already known; the defective extension is an auxiliary result outside the canonical simple-eigenvalue hypothesis. The existing partial status and general complex target are retained. The entry remains in the open count. Its difficulty and importance ratings continue to describe the surviving canonical target.

Primary reference: [complete authored PDF](#), [standalone TeX](#), **Theorem 1 and Corollary 3; Section 8 exact defective example**. [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Problem statement

Let $n \geq 1$, let $C, D, R, P \in \mathbb{C}^{n \times n}$, and write M^* for conjugate transpose. Suppose $R = R^*$, $P = P^*$, and

$$P + \lambda D^* + \lambda^{-1} D \succ 0 \quad (|\lambda| = 1).$$

For each $\eta > 0$, let X_η be the unique nonsingular solution of

$$X_\eta + (C^* + i\eta D^*)X_\eta^{-1}(C + i\eta D) = R + i\eta P$$

satisfying $\rho(X_\eta^{-1}(C + i\eta D)) < 1$, where ρ denotes spectral radius. Assume the finite limit $X_0 = \lim_{\eta \downarrow 0} X_\eta$ exists and is nonsingular. Assume the matrix polynomial

$$\mathcal{P}_0(\lambda) = \lambda^2 C^* - \lambda R + C$$

is regular, meaning $\det \mathcal{P}_0(\lambda) \not\equiv 0$. Suppose its eigenvalues on the unit circle are all simple (algebraic multiplicity one). Their number is even; denote it by $2m$.

Is it always true that

$$\text{rank}\left(\frac{X_0 - X_0^*}{2i}\right) = m?$$

Why this matters for NLA

The rank describes the non-Hermitian part of the selected matrix-equation solution in Green-function computations. It connects a limiting nonlinear solve to the spectrum of a structured quadratic polynomial.

The source proves the upper bound by m . Its earlier SIAM paper proves equality for real $C, R, P = I, D = 0$; the general complex case remains the question. Existence of the limit is assumed, not conjectured here.

References

1. C.-H. Guo, Y.-C. Kuo, W.-W. Lin, *Numerical solution of nonlinear matrix equations arising from Green's function calculations in nano research*, JCAM 236 (2012), 4166–4180: equation (1), Theorems 3 and 5, and the conjecture on p. 4172. [Author manuscript](#): pp. 1, 5–8.
2. The same authors, *On a nonlinear matrix equation arising in nano research*, SIMAX 33 (2012), 235–262, §3: the real-coefficient case.

Status check

On 2026-09-08, checked both papers' scope and Guo's [publication list through 2026](#). Searches using the JCAM title, authors, "Green rank conjecture," "weakly stabilizing," and 2025/2026 found no general resolution. The source proves only the upper bound in the stated complex setting. This is a bounded literature check.

Audit — 2026-09-10

Rechecked [Guo–Kuo–Lin, Theorem 5 and its following conjecture](#). The general complex equality remains posed, while the cited earlier real-coefficient case is settled. Exact-title, author, and rank-conjecture searches found no general resolution. The upper bound alone does not prove equality.